Post-Processing Pipeline — Collaboration Overview

1. Background & Objective

The Ego data post-processing pipeline transforms raw first-person (egocentric) video data into structured training datasets for embodied AI models. The pipeline produces **three core data modalities** from raw video input:

- **Vision** — privacy-compliant video frames and atomic task segment boundaries
- **Language** — structured text descriptions (task-level instructions and sub-task captions)
- **Action** — frame-level action data (head/camera trajectory, hand pose, retargeted robot actions)

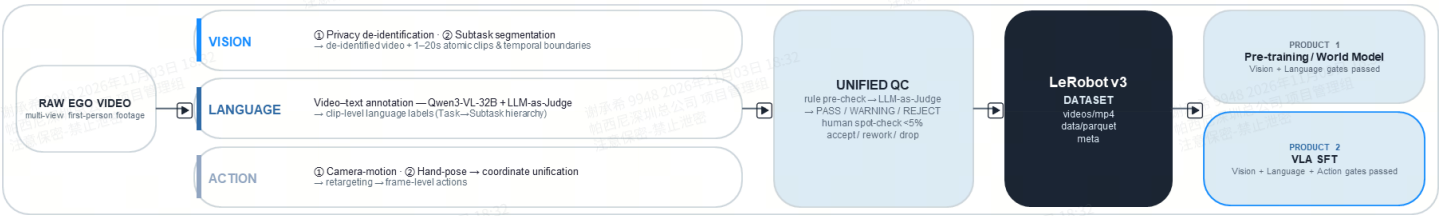
All outputs are quality-checked and packaged in **LeRobot v3 format** (video mp4 + frame-level parquet + metadata), serving two downstream use cases:

1. **Large-scale pre-training / World models** (Vision + Language)
2. **VLA imitation learning SFT** (Vision + Language + Action)

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2. Overall Architecture

The pipeline follows a modular design: **Input Layer → Vision / Language / Action Modules → Unified QC → LeRobot v3 Dataset**.



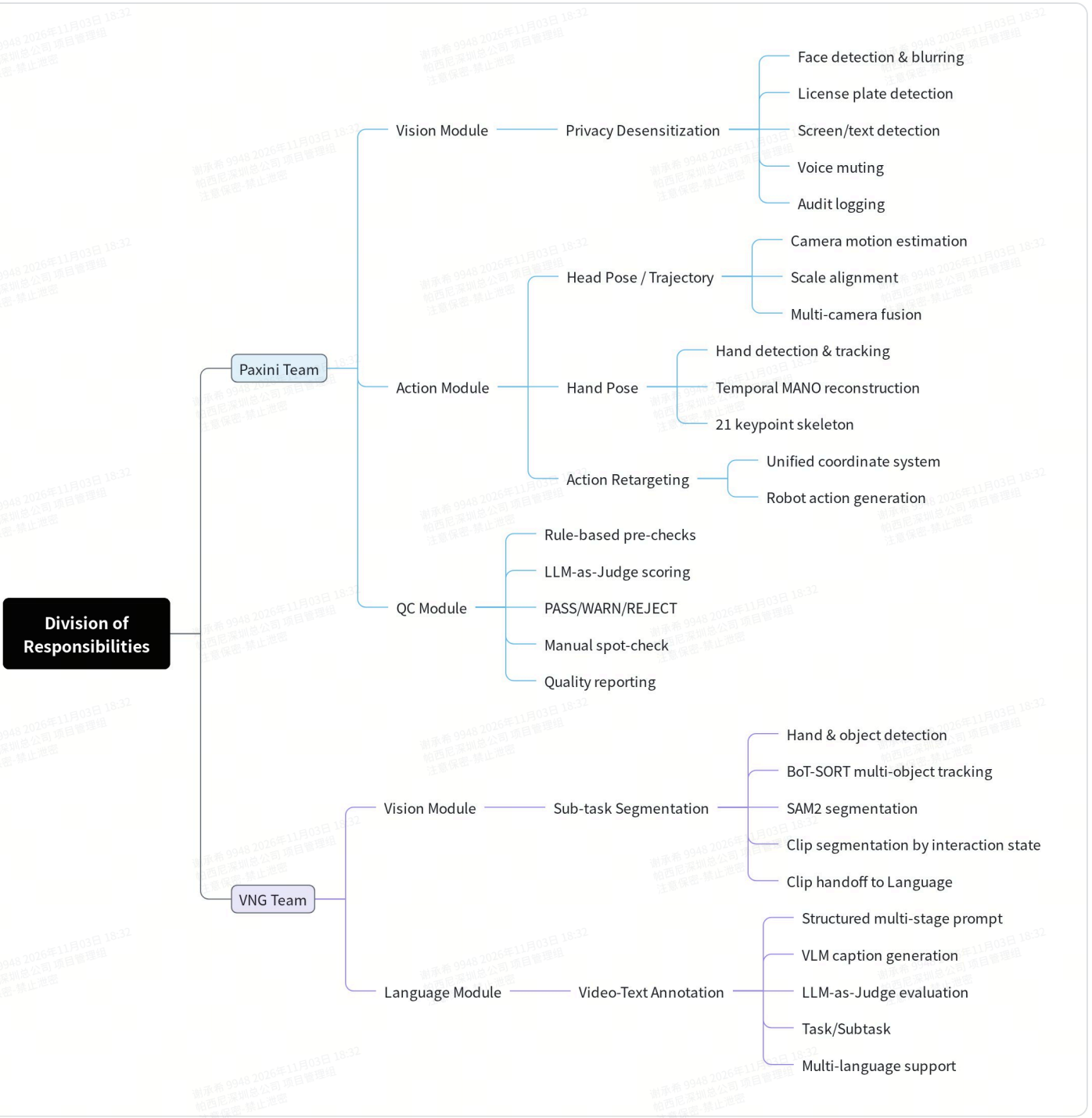
Key Module Outputs

Module	Sub-Module	Output	Serves
Vision	① de-identification ② Sub-task Segmentation	Desensitized video, atomic task clips with temporal	Both products (shared)

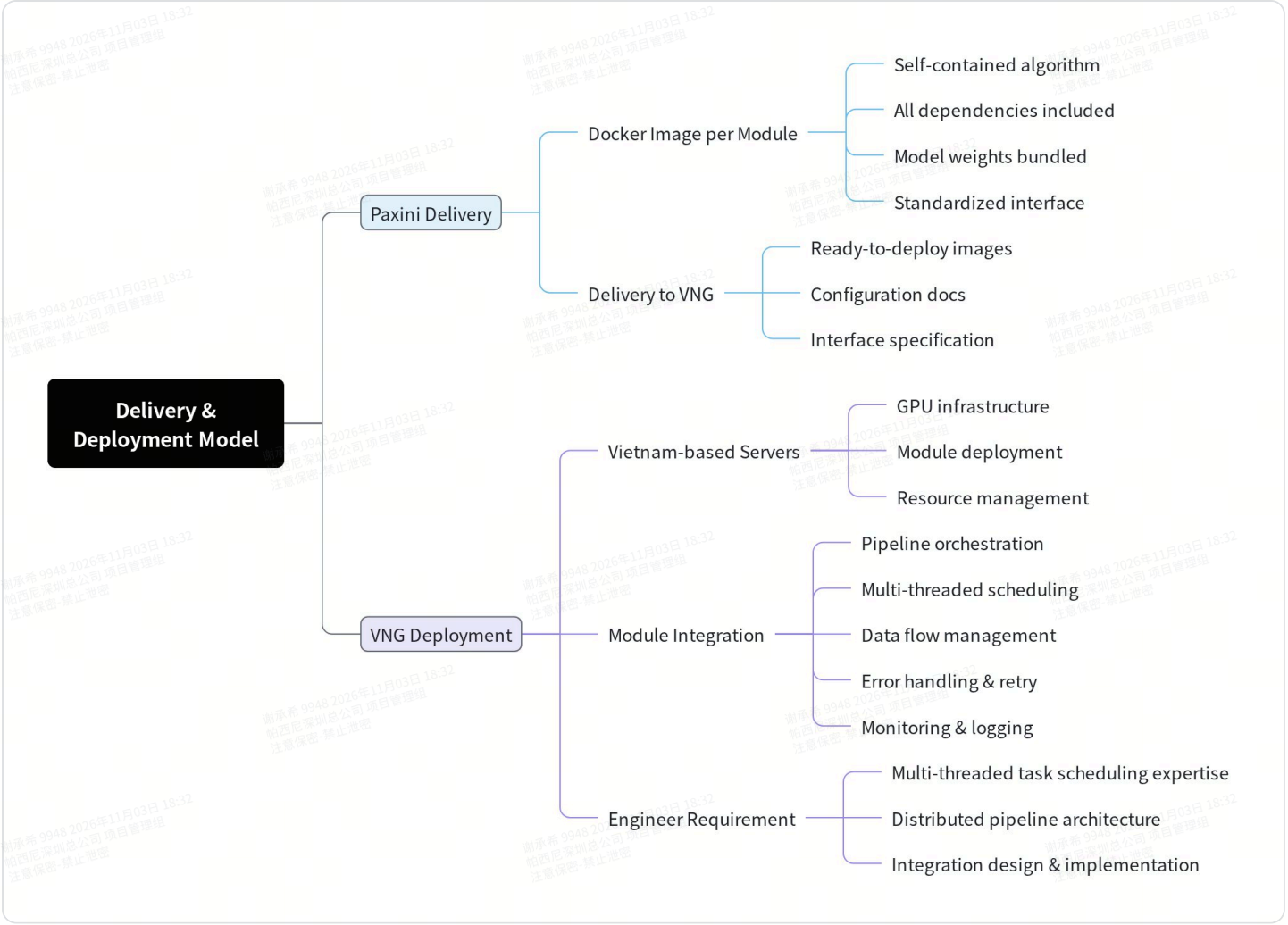
		boundaries	
Language	① Video-Text Annotation	Clip-level language labels (task instructions, sub-task descriptions)	Both products (shared)
Action	① Head Pose / Trajectory ② Hand Pose Estimation ③ Action Retargeting	Frame-level action data (head trajectory, hand pose, retargeted actions)	Primarily VLA SFT
QC	Per-module quality checks	Quality scores, grading labels, rework / rejection decisions	Gatekeeper for both products

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3. Division of Responsibilities



4. Delivery & Deployment Model



5. Interface Specification

Each Paxini-provided Docker module follows a standardized interface contract:

To be determined

Privacy De-sensitation

Item	Specification
Input	HDF5 File (video + IMU + slam)
Output	HDF5 File (preserved raw video, generated new video)

Action Mode

Item	Specification
Input	HDF5 File

Output	Generated Head/Hand Trajectory in Origin HDF5 File

VNG's integration layer is responsible for calling each module in the correct sequence, passing